

Multimodal Fusion of CLIP and ELECTRA for Interpreting Implicit Self-Harm Signals in Social Media Memes: A Human-in-the-Loop Screening Perspective

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Abstract

Self-harm-related memes present a difficult moderation problem because their meaning is often implicit and emerges from the interaction between image and text rather than from either modality alone. This study presents a multimodal binary classification approach for identifying self-harm memes by combining CLIP-based visual representations with ELECTRA-based textual representations through intermediate fusion. The work is derived from an existing end-to-end thesis study and is positioned as a screening component for moderator-facing review, not as a substitute for human judgment. The dataset was assembled from manually collected social media memes, engineered self-harm memes, and public general memes, then filtered to retain only valid multimodal samples. The final dataset comprised 2,178 memes, including 653 self-harm and 1,525 non-self-harm instances. To address labeling ambiguity in this sensitive domain, the study used ensemble pseudo-labeling with GPT-4o mini and Qwen-VL-Flash under a literature-informed annotation guideline, followed by manual adjudication for disagreements. The proposed model integrates frozen CLIP and ELECTRA encoders, projection layers, and an intermediate fusion classifier trained with tuned hyperparameters and imbalance-aware loss functions. On the validation split, the best multimodal configuration achieved 97.7% accuracy, 98.4% precision, 93.9% recall, and 96.1% F1, outperforming the strongest unimodal image-only and text-only baselines. Additional analyses showed that

balanced feature dimensions, focal loss, and simple intermediate fusion were sufficient to obtain the strongest overall performance, while a Transformer-based fusion variant offered no further gain. A separate human-reviewed ambiguity set also showed that difficult cases involving sarcasm, dark humor, and text-image conflict remain challenging, underscoring the need for human-in-the-loop review in real deployment.

Keywords: self-harm memes; multimodal screening; human-in-the-loop review; CLIP; ELECTRA; intermediate fusion; content moderation

1 Introduction

Social media has become a dominant environment for everyday communication, and memes now function as one of its most recognizable multimodal forms. At global scale, social platforms are used by billions of people, while memes circulate as replicable, culturally legible units that combine visual material, text, and shared social context [1, 2]. This communicative efficiency also makes memes suitable for conveying affectively charged or risky content in compressed, indirect, and highly shareable formats.

Within this environment, self-harm-related content warrants particular attention because online exposure may intersect with vulnerability, distress expression, and harmful normalization processes. Prior evidence has linked internet use and online self-harm content to both harmful and uncertain effects in young people, while adolescent mental health remains a major public-health concern and self-harm assessment is recognized as a sensitive clinical and policy issue [3–5]. For content governance, the difficulty is not merely the presence of explicit self-harm language or imagery. Instead, risky meaning may be embedded in humor, irony, fatalistic phrasing, symbolic objects, or emotionally charged juxtapositions that demand contextual interpretation.

This problem is especially acute for memes because their semantics are often relational. A caption may seem benign in isolation, an image may appear generic in isolation, yet their combination can imply despair, self-directed violence, or suicidal ideation. The multimodal nature of meme interpretation has been widely recognized in harmful meme research, including benchmark work showing that meme meaning frequently depends on cross-modal interaction rather than one modality alone [6, 7]. More broadly, multimodal machine learning research has long argued that fusion strategies matter because the timing and form of integration shape what kinds of cross-modal signals a system can capture [8].

These characteristics motivate the architectural choices of the present study. CLIP offers a strong visual representation backbone grounded in large-scale image-text contrastive learning [9], while ELECTRA provides an efficient contextual text encoder trained with replaced token detection and has been shown to be more sample-efficient than masked language modeling approaches of comparable size such as BERT [10, 11]. In related meme classification tasks, CLIP-based models and cross-modal interaction mechanisms have shown clear advantages for implicit and context-dependent interpretation [12, 13]. Building on that line of work, this study investigates whether

intermediate fusion of CLIP and ELECTRA can better detect implicit self-harm signals than unimodal baselines in a dataset curated for this domain.

The contributions of this article are threefold. First, it reformulates the thesis work as a moderator-oriented multimodal screening study and proposes an intermediate-fusion architecture that combines frozen CLIP image embeddings with ELECTRA text embeddings for binary self-harm meme classification. Second, it documents a weakly supervised but controlled annotation workflow for a scarce and ambiguous domain, combining ensemble pseudo-labeling with manual adjudication. Third, it analyzes the model from a system perspective relevant to *Journal on Multimodal User Interfaces*: the model is framed not as an autonomous judge of intent, but as a triage component for human-centered review workflows in which ambiguous or high-risk cases should be escalated to moderators or trained reviewers.

2 Background and Related Work

Research on harmful memes has grown rapidly, but its center of gravity has remained on hate speech, offensive content, propaganda, and related abuse categories rather than self-harm. The Hateful Memes Challenge established a widely used multimodal benchmark and made explicit how text-image interaction can defeat unimodal shortcuts [6]. Subsequent survey work similarly emphasized that harmful meme research remains unevenly distributed across categories and that domains such as self-harm are underexplored in part because suitable labeled datasets are scarce [7]. This gap is not only a data problem. It is also an interpretive problem, since self-harm memes may encode distress through dark humor, symbolic framing, or emotionally indirect language that is harder to operationalize than explicit hateful slurs or direct attacks.

The methodological literature on multimodal learning offers a useful lens for this problem. Multimodal architectures differ in where they combine information, and intermediate fusion is especially appealing when each modality must first be encoded into a meaningful latent representation before cross-modal dependencies can be exploited [8]. In harmful meme analysis, this has led to systems that go beyond independent image and text processing. MOMENTA, for example, integrates global and local multimodal cues to improve harmful meme understanding [14]. Hate-CLIPper further showed that explicit interactions among CLIP-derived representations can improve hateful meme classification [12]. More recent CLIP-based work such as Meme-CLIP has reinforced the value of vision-language representations for multi-aspect meme interpretation in real-world settings [13]. Related affective computing work has also shown that multimodal fusion becomes particularly valuable when emotional meaning is not directly recoverable from a single input stream [15].

Even so, directly transferring these lessons to self-harm memes is not straightforward. The target phenomenon differs from hatefulness in both semantics and practical consequence. Self-harm signals may be clinically adjacent, socially performative, metaphorical, or ambivalent, and they often blur the boundary between risk expression, humor, and general negativity. That ambiguity makes annotation itself a central technical issue. Weak supervision and data programming offer one principled way to expand labeled data when expert annotation is limited or expensive [16]. At the

same time, sensitive content classification studies increasingly emphasize that moderation categories are heterogeneous, context-dependent, and operationally difficult to define at dataset scale [17]. For this reason, annotation protocols and adjudication rules become part of the system contribution, not merely a preprocessing detail.

A second issue concerns the role of automation in moderation and screening. In high-stakes settings, the appropriate target is rarely fully autonomous decision-making. Work on AI-human deferral highlights the practical importance of systems that know when to hand decisions to people rather than force a hard automated classification in every case [18]. In content moderation contexts, machine-generated explanations may also help human reviewers, although their utility depends on reliability and task design [19]. For multimodal large-model safety, rationale-aware prompting frameworks such as RapGuard have further shown that structured prompting can improve analysis of text-image risk by requiring explicit attention to visual content, textual content, and their relation [20]. Taken together, these strands suggest a research niche for self-harm meme screening that is multimodal, workflow-aware, annotation-conscious, and cautious about automation.

This study occupies that niche. It does not claim clinical diagnosis, user intent inference, or policy automation. Instead, it contributes a focused multimodal screening model and an accompanying annotation-and-review pipeline for a domain in which public benchmarks remain limited, semantics are deeply relational, and deployment value depends on supporting human moderators rather than replacing them.

3 Results

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

4 This is an example for first level head—section head

4.1 This is an example for second level head—subsection head

4.1.1 This is an example for third level head—subsubsection head

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

5 Equations

Equations in L^AT_EX can either be inline or on-a-line by itself (“display equations”). For inline equations use the $\dots$ commands. E.g.: The equation $H\psi = E\psi$ is written via the command `\$H \psi = E \psi\$`.

For display equations (with auto generated equation numbers) one can use the `equation` or `align` environments:

$$\|\tilde{X}(k)\|^2 \leq \frac{\sum_{i=1}^p \|\tilde{Y}_i(k)\|^2 + \sum_{j=1}^q \|\tilde{Z}_j(k)\|^2}{p+q}. \quad (1)$$

where,

$$\begin{aligned} D_\mu &= \partial_\mu - ig \frac{\lambda^a}{2} A_\mu^a \\ F_{\mu\nu}^a &= \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc} A_\mu^b A_\nu^c \end{aligned} \quad (2)$$

Notice the use of `\nonumber` in the `align` environment at the end of each line, except the last, so as not to produce equation numbers on lines where no equation numbers are required. The `\label{}` command should only be used at the last line of an `align` environment where `\nonumber` is not used.

$$Y_\infty = \left(\frac{m}{\text{GeV}}\right)^{-3} \left[1 + \frac{3\ln(m/\text{GeV})}{15} + \frac{\ln(c_2/5)}{15}\right] \quad (3)$$

The class file also supports the use of `\mathbb{R}`, `\mathscr{R}` and `\mathcal{R}` commands. As such `\mathbb{R}`, `\mathscr{R}` and `\mathcal{R}` produces $\mathbb{R}$, $\mathscr{R}$ and $\mathcal{R}$ respectively (refer Subsubsection 4.1.1).

6 Tables

Tables can be inserted via the normal `table` and `tabular` environment. To put footnotes inside tables you should use `\footnotetext[]{\dots}` tag. The footnote appears just below the table itself (refer Tables 1 and 2). For the corresponding footnotemark use `\footnotemark[\dots]`

Table 1 Caption text

Column 1	Column 2	Column 3	Column 4
row 1	data 1	data 2	data 3
row 2	data 4	data 5 ¹	data 6
row 3	data 7	data 8	data 9 ²

Source: This is an example of table footnote. This is an example of table footnote.

¹Example for a first table footnote. This is an example of table footnote.

²Example for a second table footnote. This is an example of table footnote.

The input format for the above table is as follows:

```
\begin{table}[<placement-specifier>]
\caption{<table-caption>}\label{<table-label>}%
\begin{tabular}{@{}llll@{}}
\toprule
Column 1 & Column 2 & Column 3 & Column 4\\
\midrule
row 1 & data 1 & data 2 & data 3 \\
row 2 & data 4 & data 5\footnotemark[1] & data 6 \\
row 3 & data 7 & data 8 & data 9\footnotemark[2]\\
\botrule
\end{tabular}
\footnotetext{Source: This is an example of table footnote.
This is an example of table footnote.}
\footnotetext[1]{Example for a first table footnote.
This is an example of table footnote.}
\footnotetext[2]{Example for a second table footnote.
This is an example of table footnote.}
\end{table}
```

Table 2 Example of a lengthy table which is set to full textwidth

Project	Element 1 ¹			Element 2 ²		
	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 ± 12	780 A	1166	1239 ± 100
Element 4	500 A	961	922 ± 10	900 A	1268	1092 ± 40

Note: This is an example of table footnote. This is an example of table footnote this is an example of table footnote this is an example of table footnote this is an example of table footnote.

¹Example for a first table footnote.

²Example for a second table footnote.

In case of double column layout, tables which do not fit in single column width should be set to full text width. For this, you need to use `\begin{table*} ... \end{table*}` instead of `\begin{table} ... \end{table}` environment. Lengthy tables which do not fit in textwidth should be set as rotated table. For this, you need to use `\begin{sidewaystable} ... \end{sidewaystable}` instead of `\begin{table*} ... \end{table*}` environment. This environment puts tables rotated to single column width. For tables rotated to double column width, use `\begin{sidewaystable*} ... \end{sidewaystable*}`.

Table 3 Tables which are too long to fit, should be written using the “sidewaystable” environment as shown here

Projectile	Element 1 ¹			Element ²		
	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 $\pm$ 12	780 A	1166	1239 $\pm$ 100
Element 4	500 A	961	922 $\pm$ 10	900 A	1268	1092 $\pm$ 40
Element 5	990 A	1168	1547 $\pm$ 12	780 A	1166	1239 $\pm$ 100
Element 6	500 A	961	922 $\pm$ 10	900 A	1268	1092 $\pm$ 40

Note: This is an example of table footnote this is an example of table footnote this is an example of table footnote
this is an example of table footnote.

¹ This is an example of table footnote.

7 Figures

As per the $\text{\LaTeX}$ standards you need to use eps images for $\text{\LaTeX}$ compilation and pdf/jpg/png images for PDF $\text{\LaTeX}$ compilation. This is one of the major difference between $\text{\LaTeX}$ and PDF $\text{\LaTeX}$. Each image should be from a single input .eps/vector image file. Avoid using subfigures. The command for inserting images for $\text{\LaTeX}$ and PDF $\text{\LaTeX}$ can be generalized. The package used to insert images in $\text{\LaTeX}$ /PDF $\text{\LaTeX}$ is the graphicx package. Figures can be inserted via the normal figure environment as shown in the below example:

```
\begin{figure}[<placement-specifier>]
\centering
\includegraphics{<eps-file>}
\caption{<figure-caption>}\label{<figure-label>}
\end{figure}
```



Fig. 1 This is a widefig. This is an example of long caption this is an example of long caption this is an example of long caption this is an example of long caption

In case of double column layout, the above format puts figure captions/images to single column width. To get spanned images, we need to provide `\begin{figure*}` ... `\end{figure*}`.

For sample purpose, we have included the width of images in the optional argument of `\includegraphics` tag. Please ignore this.

8 Algorithms, Program codes and Listings

Packages `algorithm`, `algorithmicx` and `algpseudocode` are used for setting algorithms in $\text{\LaTeX}$ using the format:

```
\begin{algorithm}
\caption{<alg-caption>}\label{<alg-label>}
\begin{algorithmic}[1]
. . .
\end{algorithmic}
\end{algorithm}
```


You may refer above listed package documentations for more details before setting `algorithm` environment. For program codes, the “verbatim” package is required and the command to be used is `\begin{verbatim} ... \end{verbatim}`.

Similarly, for listings, use the `listings` package. `\begin{lstlisting} ... \end{lstlisting}` is used to set environments similar to `verbatim` environment. Refer to the `lstlisting` package documentation for more details.

A fast exponentiation procedure:

```
begin
  for i := 1 to 10 step 1 do
    expt(2,i);
    newline() od
where
proc expt(x,n) ≡
  z := 1;
  do if n = 0 then exit fi;
  do if odd(n) then exit fi;
    comment: This is a comment statement;
    n := n/2; x := x * x od;
  { n > 0 };
  n := n - 1; z := z * x od;
  print(z).
end
```

Comments will be set flush to the right margin

Algorithm 1 Calculate $y = x^n$

Require: $n \geq 0 \vee x \neq 0$

Ensure: $y = x^n$

```
1:  $y \leftarrow 1$ 
2: if  $n < 0$  then
3:    $X \leftarrow 1/x$ 
4:    $N \leftarrow -n$ 
5: else
6:    $X \leftarrow x$ 
7:    $N \leftarrow n$ 
8: end if
9: while  $N \neq 0$  do
10:  if  $N$  is even then
11:     $X \leftarrow X \times X$ 
12:     $N \leftarrow N/2$ 
13:  else [ $N$  is odd]
14:     $y \leftarrow y \times X$ 
15:     $N \leftarrow N - 1$ 
16:  end if
17: end while
```

```

for i:=maxint to 0 do
begin
  { do nothing }
end;
Write( 'Case-insensitive-');
Write( 'Pascal-keywords.' );

```

9 Cross referencing

Environments such as figure, table, equation and align can have a label declared via the `\label{#label}` command. For figures and table environments use the `\label{}` command inside or just below the `\caption{}` command. You can then use the `\ref{#label}` command to cross-reference them. As an example, consider the label declared for Figure 1 which is `\label{fig1}`. To cross-reference it, use the command `\ref{fig1}`, for which it comes up as “Figure 1”.

To reference line numbers in an algorithm, consider the label declared for the line number 2 of Algorithm 1 is `\label{algn2}`. To cross-reference it, use the command `\ref{algn2}` for which it comes up as line 2 of Algorithm 1.

9.1 Details on reference citations

Standard L^AT_EX permits only numerical citations. To support both numerical and author-year citations this template uses `natbib` L^AT_EX package. For style guidance please refer to the template user manual.

Here is an example for `\cite{...}`: [1]. Another example for `\citep{...}`: [2]. For author-year citation mode, `\cite{...}` prints Jones et al. (1990) and `\citep{...}` prints (Jones et al., 1990).

All cited bib entries are printed at the end of this article: [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] and [13].

10 Examples for theorem like environments

For theorem like environments, we require `amsthm` package. There are three types of predefined theorem styles exists—`thmstyleone`, `thmstyletwo` and `thmstylethree`

<code>thmstyleone</code>	Numbered, theorem head in bold font and theorem text in italic style
<code>thmstyletwo</code>	Numbered, theorem head in roman font and theorem text in italic style
<code>thmstylethree</code>	Numbered, theorem head in bold font and theorem text in roman style

For mathematics journals, theorem styles can be included as shown in the following examples:

Theorem 1 (Theorem subhead) *Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Proposition 2 *Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Example 1 Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Remark 1 Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Definition 1 (Definition sub head) Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text.

Additionally a predefined “proof” environment is available: `\begin{proof}` ... `\end{proof}`. This prints a “Proof” head in italic font style and the “body text” in roman font style with an open square at the end of each proof environment.

Proof Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. □

12 Discussion

Discussions should be brief and focused. In some disciplines use of Discussion or ‘Conclusion’ is interchangeable. It is not mandatory to use both. Some journals prefer a section ‘Results and Discussion’ followed by a section ‘Conclusion’. Please refer to Journal-level guidance for any specific requirements.

13 Conclusion

Conclusions may be used to restate your hypothesis or research question, restate your major findings, explain the relevance and the added value of your work, highlight any limitations of your study, describe future directions for research and recommendations.

In some disciplines use of Discussion or ‘Conclusion’ is interchangeable. It is not mandatory to use both. Please refer to Journal-level guidance for any specific requirements.

Supplementary information. If your article has accompanying supplementary file/s please state so here.

Authors reporting data from electrophoretic gels and blots should supply the full unprocessed scans for key as part of their Supplementary information. This may be requested by the editorial team/s if it is missing.

Please refer to Journal-level guidance for any specific requirements.

Acknowledgements. Acknowledgements are not compulsory. Where included they should be brief. Grant or contribution numbers may be acknowledged.

Please refer to Journal-level guidance for any specific requirements.

Declarations

Some journals require declarations to be submitted in a standardised format. Please check the Instructions for Authors of the journal to which you are submitting to see if you need to complete this section. If yes, your manuscript must contain the following sections under the heading ‘Declarations’:

- Funding
- Conflict of interest/Competing interests (check journal-specific guidelines for which heading to use)
- Ethics approval and consent to participate
- Consent for publication
- Data availability
- Materials availability
- Code availability
- Author contribution

If any of the sections are not relevant to your manuscript, please include the heading and write ‘Not applicable’ for that section.

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Scientific Reports: <https://www.nature.com/srep/journal-policies/editorial-policies>

BMC journals: <https://www.biomedcentral.com/getpublished/editorial-policies>

Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

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